

Distance from a point to a line



- 1 Write down the formula to find the perpendicular distance d from the point (x_1, y_1) to the line $Ax + By + C = 0$.
 - 2 Find the perpendicular distance from the following points to the given line:
 - a Point $(0, 0)$, Line : $4x + 7y + 5 = 0$
 - b Point $(-9, -3)$, Line : $6x + 3y + 1 = 0$
 - c Point $(3, 5)$, Line : $4x + 3y + 1 = 0$
 - d Point $(4, -3)$, Line : $24x + 7y - 3 = 0$
 - e Point $(5, 4)$, Line : $5x - 12y = 0$
 - f Point $(-5, 3)$, Line : $3x - 2y - 5 = 0$
 - 3 Find the perpendicular distance from the point $(8, 7)$ to the line $9x - 9y - 0 = 0$.
 - 4 Find the value of k with the following conditions:
 - a The perpendicular distance from the point $(-5, -1)$ to the line $-2x - 7y + k = 0$ is $\frac{10}{\sqrt{53}}$.
 - b The perpendicular distance from the point $((6, k)$, to the line $6x + (-2y) + (-6) = 0$ is $\frac{40}{\sqrt{40}}$.
 - 5 Consider the lines $L_1: x + 2y + 5 = 0$ and $L_2: x + by + 20 = 0$.
 - a Find the distance between L_1 and the origin.
 - b Solve for the value(s) of b such that L_1 is twice as far from the origin as L_2 .
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Geometric applications

- 6 Consider the circle represented by the equation $x^2 + y^2 = 0.81$.
 - a Find the centre of the circle.
 - b Find the radius of the circle.
 - c Find the perpendicular distance from the origin to the line $-4x + (-7y) + 8 = 0$.
 - d State the number of intersections between the line and circle. Explain your answer.

- 7 Consider the circle represented by the equation $(x - (-3))^2 + (y - (-2))^2 = 4$.
- Find the centre of the circle.
 - Find the radius of the circle.
 - Find the perpendicular distance from the centre of the circle to the line $-3x + 5y + (-5) = 0$.
 - State the number of intersections between the line and circle. Explain your answer.
- 8 The three points $A(-3, -4)$, $B(0, -10)$ and $C(-6, -7)$ form a triangle.
- Find the gradient of the line AC .
 - Find the equation of the line AC in general form.
 - Find the midpoint, D , of the line AC .
 - Find the gradient of the line BD .
 - Is BD perpendicular to AC ?
 - What type of triangle is ABC ?
 - Find the length of the line BD . Give your answer in simplest surd form.
 - Find the length of the line AC . Give your answer in simplest surd form.
 - Find the area of triangle ABC .
 - Find the coordinates of the point E that would make $ABCE$ a parallelogram.
- 9 The points $A(5, 7)$, $B(9, 10)$, $C(6, 4)$ and $D(2, 1)$ are the vertices of a parallelogram.
- Determine the equation of the line going through AB .
 - If AB is the length of the parallelogram, find the perpendicular height of the parallelogram.
 - Determine the exact area of the parallelogram.